Online Appendix to "A new algorithm for sign restrictions in vector autoregressions"

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A Technical Appendix

A.1 Full Gibbs sampler for VARs with sign restrictions

The Gibbs sampler presented in the main paper is a quite accurate description of the core algorithmic steps required in order to estimate the VAR model with factor structure in the residuals. Nevertheless, in order to produce empirical and other results, we have relied on the hierarchical horseshoe prior of Carvalho et al. (2010), two fast algorithms from drawing from the Normal (Bhattacharya et al., 2016) and truncated Normal (Botev, 2017) distributions, respectively, and the slice sampler of Neal (2003) in order to update the horseshoe prior parameters. Therefore, it is important to rewrite the Gibbs sampling algorithm in full, and give further explanations about the three enhancements that guarantee a fast and reliable algorithm in high dimensions.

I repeat the full prior specification, which now includes the hierarchical horseshoe prior on $\boldsymbol{\phi}_i$. The priors for the ith VAR equation, $i = 1, ..., n$ is:

$$\boldsymbol{\phi}_i|\sigma_i^2, \tau_i^2, \boldsymbol{\Psi}_i^2 \sim N_k\left(\mathbf{0}, \sigma_i^2\tau_i^2\boldsymbol{\Psi}_i^2\right), \quad \boldsymbol{\Psi}_i^2 = diag\left(\psi_{i,1}^2, ..., \psi_{i,k}^2\right), \tag{A.1}$$

$$\psi_{i,j} \sim Cauchy^+(0,1), \quad j = 1, ..., k, \tag{A.2}$$

$$\tau_i \sim Cauchy^+(0,1), \tag{A.3}$$

$$\boldsymbol{\Lambda}_{ij} \sim \begin{cases} N\left(0, \underline{h}_{ij}\right) I(\Lambda_{ij} > 0), & \text{if } S_{ij} = 1, \\ N\left(0, \underline{h}_{ij}\right) I(\Lambda_{ij} < 0), & \text{if } S_{ij} = -1, \\ \delta_0\left(\boldsymbol{\Lambda}_{ij}\right), & \text{if } S_{ij} = 0, \\ N\left(0, \underline{h}_{ij}\right), & \text{otherwise,} \end{cases} \quad j = 1, ..., r, \tag{A.4}$$

$$\mathbf{f}_t \sim N_r(\mathbf{0}, \mathbf{I}), \tag{A.5}$$

$$\sigma_i^2 \sim inv - Gamma\left(\underline{\rho}_i, \underline{\kappa}_i\right), \tag{A.6}$$

where we set $\underline{h}_{ij} = 4$, $\underline{\rho}_i = 1$ and $\underline{\kappa}_i = 0.01$.

Under these priors, the full factor sign restrictions algorithm takes the following form

Factor sign restrictions (FSR) algorithm

1. Sample $\boldsymbol{\phi}_i$ for $i = 1, ..., n$ from

$$\boldsymbol{\phi}_i | \boldsymbol{\Sigma}, \boldsymbol{\Lambda}, \mathbf{f}, \mathbf{y} \sim N_k \left(\overline{\mathbf{V}}_i \left(\sum_{t=1}^{T} \sigma_i^{-2} \mathbf{x}_t' \widetilde{\mathbf{y}}_{it} \right), \overline{\mathbf{V}}_i \right), \tag{A.7}$$

   where $\widetilde{\mathbf{y}}_{it} = \mathbf{y}_{it} - \boldsymbol{\Lambda}_i \mathbf{f}_t$ and $\overline{\mathbf{V}}_i^{-1} = \left(\underline{\mathbf{V}}_i^{-1} + \sum_{t=1}^{T} \sigma_i^{-2} \mathbf{x}_t' \mathbf{x}_t \right)$. We use the efficient sampler of Bhattacharya et al. (2016) in order to sample these elements.

2. Sample ψ_{ij} using slice sampling (Neal, 2003)

   a. Set $\eta_{ij} = 1/\psi_{ij}^2$ using the last available sample of ψ_{ij}^2.

   b. Sample a random variable u from

$$u | \eta_{ij} \sim Uniform \left(0, \frac{1}{1 + \eta_{ij}} \right). \tag{A.8}$$

   c. Sample η_{ij} from

$$\eta_{ij} \sim e^{\frac{\phi_{ij}^2}{2\sigma_i^2} \eta_{ij}} I \left(\frac{u}{1-u} > \eta_{ij} \right) \tag{A.9}$$

   and set $\psi_{ij} = 1/\sqrt{\eta_{ij}}$.

3. Sample τ_i using slice sampling (Neal, 2003)

   a. Set $\xi_i = 1/\tau_i^2$ using the last available sample of τ_i^2.

   b. Sample a random variable u from

$$v | \xi_{ij} \sim Uniform \left(0, \frac{1}{1 + \xi_{ij}} \right). \tag{A.10}$$

   c. Sample ξ_i from

$$\xi_i \sim \gamma \left((k+1)/2, v \frac{2\sigma^2}{\sum \left(\frac{\phi_{ij}}{\psi_{ij}} \right)^2} \right), \tag{A.11}$$

   where $\gamma(\bullet)$ is the lower incomplete gamma function, and set $\tau_i = 1/\sqrt{\xi_i}$

4. Sample $\mathbf{\Lambda}_{ij}$ from univariate conditional posteriors (Geweke, 1996) of the form

$$\mathbf{\Lambda}_{ij}|\mathbf{\Lambda}_{-ij}\mathbf{\Phi},\mathbf{\Sigma},\mathbf{f},\mathbf{y}\sim TN_{(\mathbf{a}_{ij},\mathbf{b}_{ij})}\left(\overline{\lambda}_{ij}-\overline{h}_{ij}\sum_{l\neq j}\overline{h}_{il}^{-1}\left(\mathbf{\Lambda}_{il}-\overline{\lambda}_{il}\right),\overline{h}_{ij}\right), \tag{A.12}$$

where $\overline{\lambda}_{ij}$ and $\overline{h}_{ij}$ denote the ij^{th} elements of the joint posterior mean and variance, respectively, of $\mathbf{\Lambda}_i$. The joint posterior variance is $\overline{\mathbf{H}}^{-1}=\left(\underline{\mathbf{H}}^{-1}+\sum_{t=1}^{T}\sigma_i^{-2}\mathbf{f}_t'\mathbf{f}_t\right)$ and the joint posterior mean is $\overline{\mathbf{H}}\left(\sum_{t=1}^{T}\sigma_i^{-2}\mathbf{f}_t'\widehat{\mathbf{y}}_{it}\right)$ with $\widehat{\mathbf{y}}_{it}\equiv\boldsymbol{\varepsilon}_{it}=\mathbf{y}_{it}-\phi_i\mathbf{x}_t$. Here $TN_{(\mathbf{a}_{ij},\mathbf{b}_{ij})}(\bullet)$ denotes the **univariate** truncated Normal distribution with bounds:

$$(\mathbf{a}_{ij},\mathbf{b}_{ij})=\begin{cases}(-\infty,0) & \text{if } S_{ij}=-1,\\ (0,\infty) & \text{if } S_{ij}=1,\\ (0,0) & \text{if } S_{ij}=0,\\ (-\infty,\infty) & \text{otherwise,}\end{cases} \tag{A.13}$$

We use the efficient univariate truncated Normal generator provided by Botev (2017) in order to sample these elements.

5. Sample $\mathbf{f}_t$ for $t=1,...,T$ from

$$\mathbf{f}_t|\mathbf{\Lambda},\mathbf{\Sigma},\mathbf{\Phi},\mathbf{y}\sim N\left(\overline{\mathbf{G}}\left(\mathbf{\Lambda}\mathbf{\Sigma}^{-1}\widehat{\mathbf{y}}_t\right),\overline{\mathbf{G}}\right), \tag{A.14}$$

where $\overline{\mathbf{G}}^{-1}=(\mathbf{I}_r+\mathbf{\Lambda}'\mathbf{\Sigma}\mathbf{\Lambda})$. Post-process the draws of the $T\times r$ matrix $\mathbf{f}=(\mathbf{f}_1,...,\mathbf{f}_T)'$ such that its r columns (corresponding to structural shocks) are uncorrelated and standardized to unit variance. This is done by applying first the Gram-Schmidt procedure and subsequently dividing each column of $\mathbf{f}$ with its standard deviation.

6. Sample σ_i^2 for $i=1,...,n$ from

$$\sigma_i^2|\mathbf{\Lambda},\mathbf{f},\mathbf{\Phi},\mathbf{y}\sim inv-Gamma\left(\frac{T}{2}+\underline{\rho}_i,\left[\underline{\kappa}_i^{-1}+\sum_{t=1}^{T}\left(\mathbf{y}_{it}-\phi_i\mathbf{x}_t-\mathbf{\Lambda}_i\mathbf{f}_t\right)'\left(\mathbf{y}_{it}-\phi_i\mathbf{x}_t-\mathbf{\Lambda}_i\mathbf{f}_t\right)\right]^{-1}\right) \tag{A.15}$$

A.2 Priors and efficient sampling

While the Gibbs sampler presented above is based on standard sampling steps for Bayesian VARs and static factor models (Lopes and West, 2004), I discuss here how to build an even more reliable algorithm that can handle larger models with the same ease it can estimate smaller models. The main computational concerns with the core algorithm presented above

stem from the high dimensions of $\mathbf{\Phi}$[1] and the fact that $\mathbf{\Lambda}$ is both latent and has to be sampled from a restricted (truncated) Normal distribution.

First, overall the proposed factor-VAR had disturbances that have a diagonal covariance matrix. Therefore, the parameters $\mathbf{\Phi}$ can be sampled equation-by-equation (which is what Baumeister and Hamilton, 2015 also do in the SVAR setting) using (19). Instead of having to sample one large vector of nk parameters, n independent univariate autoregressions with k parameters each can be estimated – a step that is also trivial to parallelize using modern computers. This point has been established recently in Carriero et al. (2019).

Second, I additionally follow ideas in Bhattacharya et al. (2016) and use an efficient sampler for the Normal distribution, which makes full use of the Woodbury identity in order to sample from equation (19)efficiently. The idea is that during sampling from the Normal distribution one has to obtain the Cholesky of a large matrix $(\overline{\mathbf{V}}_i)$ which requires $\mathcal{O}(k^3)$ algorithmic operations. The transformation of Bhattacharya et al. (2016) allows sampling with worst case algorithmic complexity of $\mathcal{O}(T^2k)$ operations. Therefore, gains from replacing this step with the Bhattacharya et al. (2016) approach can be very substantial as either the dimension n of the VAR or the number of lags p increase. In particular, as $k = np+1$ becomes larger than T, a significantly smaller number of algorithmic operations will be required in order to sample from the posterior of $\mathbf{\Phi}$.

Third, as the dimension of $\mathbf{\Phi}$ increases polynomially in n and p, computation is not the sole concern; regularization also becomes an important aspect of statistical inference. Traditionally, empirical Bayes priors such as the Minnesota prior, have been used both in reduced-form and structural VARs; see Giannone et al. (2015) and Baumeister and Hamilton (2015), respectively. However, little theory exists on the high-dimensional shrinkage properties and posterior consistency of such empirical Bayes rules. In contrast, the *horseshoe prior for sparse signals* proposed by Carvalho et al. (2010) has been shown to lead to Bayes estimates that are consistent a-posteriori, and that attain a risk equivalent to the (Bayes) oracle; see Armagan et al. (2013) and Ghosh et al. (2016) and references therein.[2] Using equation (15),

[1]For example, in a 50 variable VAR with 4 lags and an intercept this matrix has 10,000+ elements.

[2]van der Pas et al. (2014) also show that the horseshoe has good frequentist properties and can attain minimax-adaptive risk up to a constant, for squared error (l_2) loss. Recently, Kowal et al. (2019) also establish the excellent shrinkage properties of the horseshoe in a dynamic linear model for time-series data.

I specify the prior covariance matrix to have the following hierarchical structure

$$\begin{aligned}
\boldsymbol{\phi}_i|\sigma_i^2\tau_i^2\boldsymbol{\Psi}_i &\sim N_k\left(\mathbf{0},\underline{\mathbf{V}}_i\right), && \text{(A.16)}\\
\underline{\mathbf{V}}_{i,(jj)} &= \sigma_i^2\tau_i^2\psi_{i,j}^2, && \text{(A.17)}\\
\psi_{i,j} &\sim Cauchy^{+}\left(0,1\right), && \text{(A.18)}\\
\tau_i &\sim Cauchy^{+}\left(0,1\right), && \text{(A.19)}
\end{aligned}$$

where $\underline{\mathbf{V}}_{i,(jj)}$ denotes the jth diagonal element of prior covariance matrix $\underline{\mathbf{V}}_i$, $\boldsymbol{\Psi}_i = diag\left(\psi_{i,1}^2,...,\psi_{i,k}^2\right)$, and $Cauchy^{+}$ denotes the half-Cauchy distribution with support on the set of positive real numbers $\mathbb{R}^{+}$.[3] The most important aspect of the horseshoe prior is that it requires absolutely no input from the researcher, while retaining at the same time its excellent shrinkage properties. There are various reparametrizations of this prior and associated MCMC sampling schemes. Here I follow Neal (2003) and use a slice sampler (within the Gibbs sampler algorithm) that allows to update $\psi_{i,(jj)}$ and τ_i efficiently in high-dimensions.

Fourth, a major challenge in the proposed algorithm is that in Step 2, $\boldsymbol{\Lambda}_{ij}$ has to be sampled from a multivariate truncated Normal distribution which is notoriously hard to simulate from (Geweke, 1996). In order to deal with this significant computational challenge, I follow Geweke (1996) and sample each $\boldsymbol{\Lambda}_{ij}$ conditional on all remaining $\boldsymbol{\Lambda}_{-ij}$ elements. Most importantly, I adopt the exact minimax tilting method for generating i.i.d. data from a truncated normal distribution that was recently proposed by Botev (2017). This method allows orders of magnitude computational improvements relative to the truncated normal sampler proposed by Geweke (1996) and others; see Botev (2017) for a review of this literature. Given absence of prior information on the variance of $\boldsymbol{\Lambda}$ (above and beyond the information we have on the sign restrictions) I set $\underline{h}_{ij} = 4$ in the remainder of the paper, which is a noninformative choice for the parameters of a loadings matrix.

All the considerations above ensure that the algorithm proposed in the previous subsection is not only fast, but is also numerically stable and can scale up to much larger VAR dimensions than ever considered before in the literature. Note that there are remaining prior settings and algorithmic steps (those involving parameters $\mathbf{f}_t$ and $\boldsymbol{\Sigma}$), but these steps are already computationally trivial and may not be made more efficient. Additionally, the prior for $\mathbf{f}_t$ is fixed by the required identification restrictions, and the $\boldsymbol{\Sigma}$ are integrated out with fairly noninformative values (I set $\rho_i = 1$ and $\kappa_i = 0.01$ for all i, in all VARs estimated in this

[3]Note that the prior covariance matrix of $\boldsymbol{\phi}_i$ is a function of the VAR variance σ_i^2. This is done in order to enhance numerical stability when the endogenous variables in the VAR are measured in different units, even though in practice it is trivial to specify this prior to be independent of σ_i^2 (which would be essential in case we want to specify σ_i^2 to be time-varying using, for instance, a stochastic volatility specification.

paper).

A.3 Likelihood-based testing of identifying assumptions in VARs

Our modeling assumption that the structural shocks $\mathbf{f}_t$ are latent parameters that have to be estimated from the likelihood, has an enormous implication: sign, or any other structural restrictions, become part of the model likelihood and can be explicitly tested the same way economists test other parametric restrictions in regression models (e.g. inequality constraints as in Geweke, 1996). This concept might not make sense for economic shocks that are indisputable such as the effects of an aggregate demand/supply shock, however, there are cases of shocks where the expected sign might not be known a-priori with certainty. Alternatively, a researcher might want to statistically test the plausibility of certain zero restrictions, or simply compare the performance of two different VAR models (e.g. with different number of lags and/or variables) given the same set of identifying restrictions. In the context of the proposed VAR model with factor structure, all these cases can be tested explicitly using marginal likelihoods or Bayes factors. Parametric (i.e. zero or sign) restrictions on $\mathbf{\Lambda}$ that agree with the information in the data will result in higher marginal data likelihood relative to restrictions that are not supported by the data. Put differently, given the decomposition in equation (4), plausible restrictions in $\mathbf{\Lambda}$ will result in estimation of a more precise unconditional VAR covariance matrix $\mathbf{\Omega}$.

Marginal likelihoods are not numerically stable in high-dimensional VARs (Giannone et al., 2015) and they can be demanding to compute even in smaller VARs with many layers of latent parameters (e.g. hierarchical priors like the horseshoe; stochastic variances; Markov-switching coefficients). In order to deal with this computational aspect, I instead propose to calculate the Deviance Information Criterion (DIC) of Spiegelhalter et al. (2002) as a default criterion for assessing model fit. For the matrix of VAR model parameters $\mathbf{\Theta} = \left(\mathbf{\Phi}, \mathbf{\Lambda}, \{\mathbf{f}_t\}_{t=1}^T, \mathbf{\Lambda}\right)$ the DIC is defined as the quantity

$$DIC = -4E_{p(\mathbf{\Theta}|\mathbf{y})}\left(\log f\left(\mathbf{y}|\mathbf{\Theta}\right)\right) + 2\log f\left(\mathbf{y}|\widehat{\mathbf{\Theta}}\right), \tag{A.20}$$

where $\log f\left(\mathbf{y}|\mathbf{\Theta}\right)$ is the log of the likelihood function implied by the regression in (14). The first term in the criterion above is the expectation of the likelihood w.r.t the parameter posterior, and can be obtained numerically using Monte Carlo integration by simply evaluating the likelihood at each MCMC draw from the posterior of the parameters $\mathbf{\Theta}$. The second term is the likelihood function evaluated at an estimate $\widehat{\mathbf{\Theta}}$ of high posterior density (typically posterior mean or mode). As with all other information criteria used in statistics/econometrics, lower values signify better fit. The DIC is not a first-order

approximation to the marginal likelihood, in the same way that the Bayesian Information Criterion (BIC) is. The marginal likelihood, also known as *prior predictive distribution*, addresses the issue of how well the data are predicted by the priors. In this sense, the DIC is a criterion that is closely related to measuring fit according to the *posterior predictive distribution*, rather than marginal likelihoods. As a result, for the purpose of assessing the fit of a VAR that is intended to be used for out-of-sample projections (impulse responses, forecast error variance decompositions etc), the DIC can be considered as a more appropriate predictive measure of fit compared to marginal likelihoods or alternative in-sample measures of fit.

B Data Appendix

B.1 Data for the simulation study

The data used in the simulation exercise are an augmented version of the data used in Arias et al. (2019). The measure of commodity prices they provide comes from Global Financial Data, while all remaining variables come originally from St Louis' Federal Reserve Economic Data (FRED; https://fred.stlouisfed.org/). Monthly GDP and GDP deflator are constructed using interpolation, and the reader is referred to the data supplement of Arias et al. (2019) for more information. The variables these authors use are augmented with stock prices, M1, unemployment rate, industrial production, employment, consumer prices (total), core consumer prices (total, less food and energy) and personal consumption deflator.

All variable mnemonics, short descriptions, and sources are visible in Table B1. The last column (Tcode) refers to the stationarity transformation used for each variable, namely 1: levels, and 5: first differences of logarithm (growth rates).

Table B1: Data used in Monte Carlo exercise

Mnemonic	Description	Source	Tcode
GDPC1	Monthly real GDP	Arias et al. (2019)	5
GDPDEFL	Monthly GDP deflator	Arias et al. (2019)	5
FEDFUNDS	Fed funds rate	Arias et al. (2019)	1
CPRINDEX	Commodity price index	Arias et al. (2019)	5
TRARR	Total reserves	Arias et al. (2019)	5
BOGNONBR	Nonborrowed reserves	Arias et al. (2019)	5
ˆGSPC	S&P 500 prices	Yahoo! Finance	5
M1REAL	Real M1 money stock	FRED	5
UNRATE	Unemployment rate	FRED	1
INDPRO	Industrial production index, all industries	FRED	5
PAYEMS	Employment, total	FRED	5
CPIAUSL	Consumer price index, all items	FRED	5
CPILFESL	Core CPI	FRED	5
PCEPILFE	Core PCE deflator	FRED	5

B.2 Data used for the large-scale VAR model for the US

The full list of variables is shown in Table B2 and they pertain to the sample 1985Q1 - 2013Q2. The original mortgage rate used by Furlanetto et al. (2019) was only available after 1990Q1, and for that reason it has been replaced by the 30-year mortgage rate provided by FRED (contemporaneous correlation between the two series is 0.9967). All remaining series used in the empirical exercise are exactly those described in Table 11 of Furlanetto et al. (2019), augmented with a few additional measures of output, consumer prices, and stock prices. The fifteen variables used in the large VAR can be seen in the first column of Table 6. In this list **stock prices** refers to S&P500 while **stock prices 2** refers to Dow Jones Industrial Average. Similarly, **spread** refers to the Baa minus fed funds rate spread, while **spread 2** refers to the GZ credit spread.

Table B2: Data used in the empirical exercise

Variable	Description	Source
GDP	Log of real GNP/GDP	Federal Reserve Bank of Philadelphia
GDP deflator	Log of price index for GNP/GDP	Federal Reserve Bank of Philadelphia
Interest rate	3-month treasury bill	Federal Reserve Bank of St. Louis
Investment	Log of real gross private domestic investment	Federal Reserve Bank of St. Louis
Stock prices	Log of real S&P 500	Yahoo! Finance
Total credit	Log of loans to non-financial private sector	Board of Governors of the Federal Reserve System
Mortgages	Log of home mortgages of households and non-profit organizations	Board of Governors of the Federal Reserve System
Real estate value	Log of real estate at market value of households and non-profit organizations	Board of Governors of the Federal Reserve System
Corporate bond yield	Moody's baa corporate bond yield	Federal Reserve Bank of St. Louis
Federal funds rate	Federal funds rate	Federal Reserve Bank of St. Louis
GZ credit spread	Senior unsecured corporate bond spreads (non-financial firms)	Gilchrist and Zakrajšek (2012)[a]
EBP	Excess bond premium	Gilchrist and Zakrajšek (2012)[a]
VIX	Stock market volatility index	Bloom (2009)[b]
Mortgage rates	Home mortgages, fixed 30YR, Effective interest rate	Federal Reserve Bank of St. Louis
Employment	Log of total nonfarm employment	Federal Reserve Bank of Philadelphia
Core prices	Log of core consumer price index	Federal Reserve Bank of Philadelphia
Stock prices 2	Log of real DJIA	Yahoo! Finance

[a] Gilchrist, Simon, and Egon Zakrajšek (2012), Credit Spreads and Business Cycle Fluctuations. American Economic Review, 102 (4): 1692-1720.

[b] Bloom, Nicholas (2009), The Impact of Uncertainty Shocks. Econometrica, 77: 623-685.

C Additional results

C.1 Additional results for the first Monte Carlo exercise

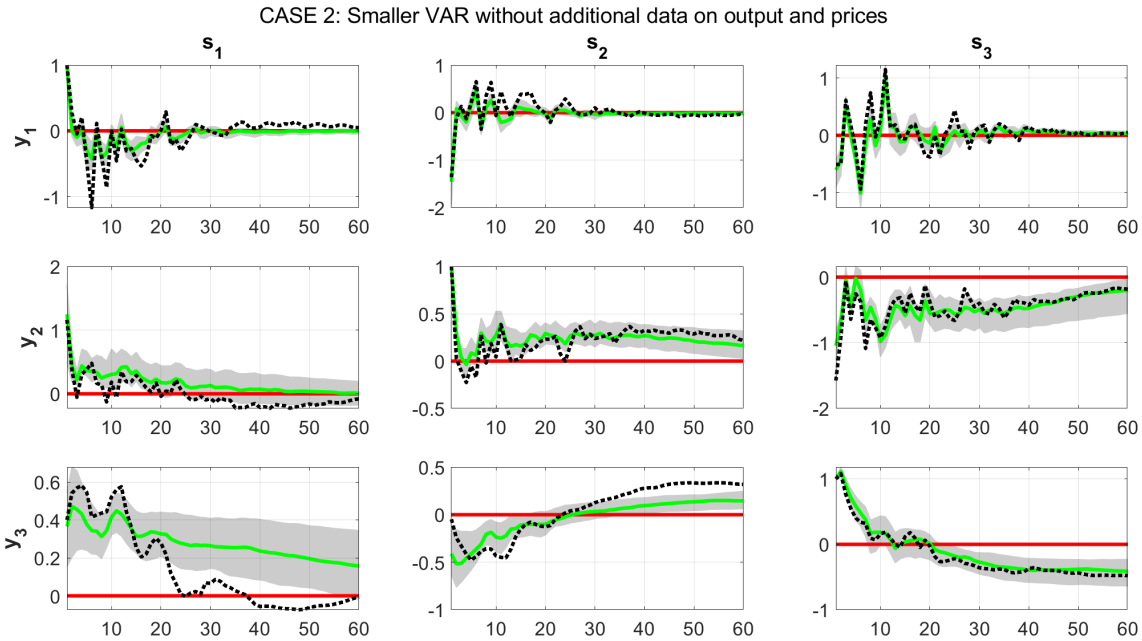


Figure C1: *Impulse response functions of the first three artificially generated variables (denoted as y_1, y_2, y_3) in response to the three identified shocks (denoted as s_1, s_2, s_3) in model C2 (misspecified VAR dimension). The green solid lines show the posterior median IRFs over the 500 Monte Carlo iterations, and the gray shaded areas their associated 90% bands. The true IRFs based on the DGP are shown using the black dashed lines.*

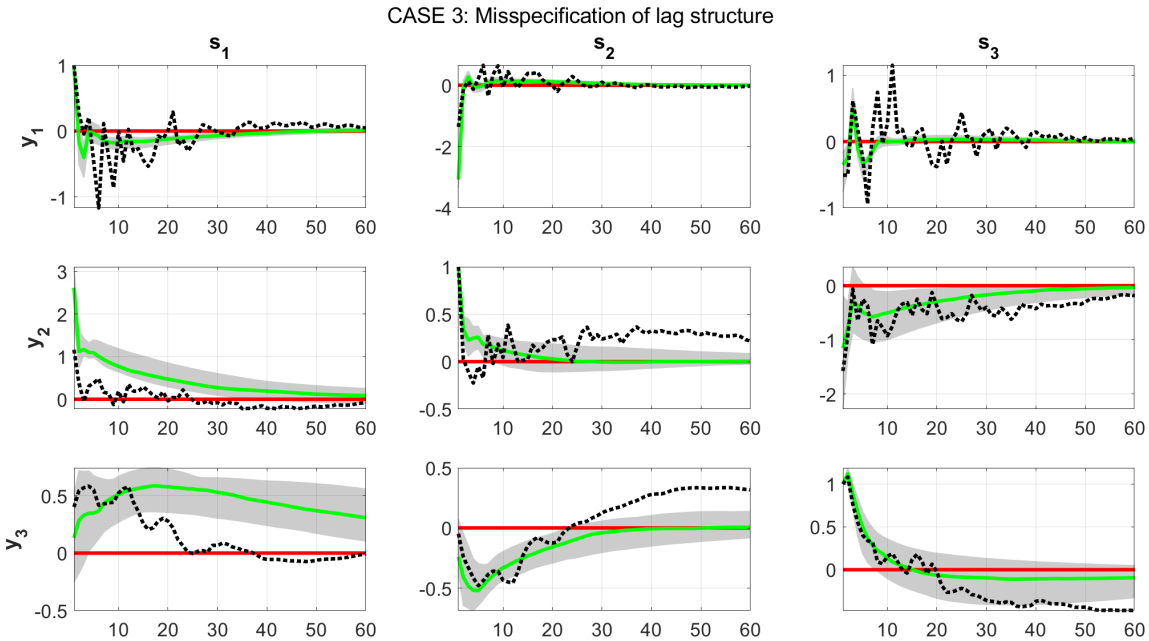


Figure C2: *Impulse response functions of the first three artificially generated variables (denoted as y_1, y_2, y_3) in response to the three identified shocks (denoted as s_1, s_2, s_3) in model C3 (misspecified number of lags). The green solid lines show the posterior median IRFs over the 500 Monte Carlo iterations, and the gray shaded areas their associated 90% bands. The true IRFs based on the DGP are shown using the black dashed lines.*

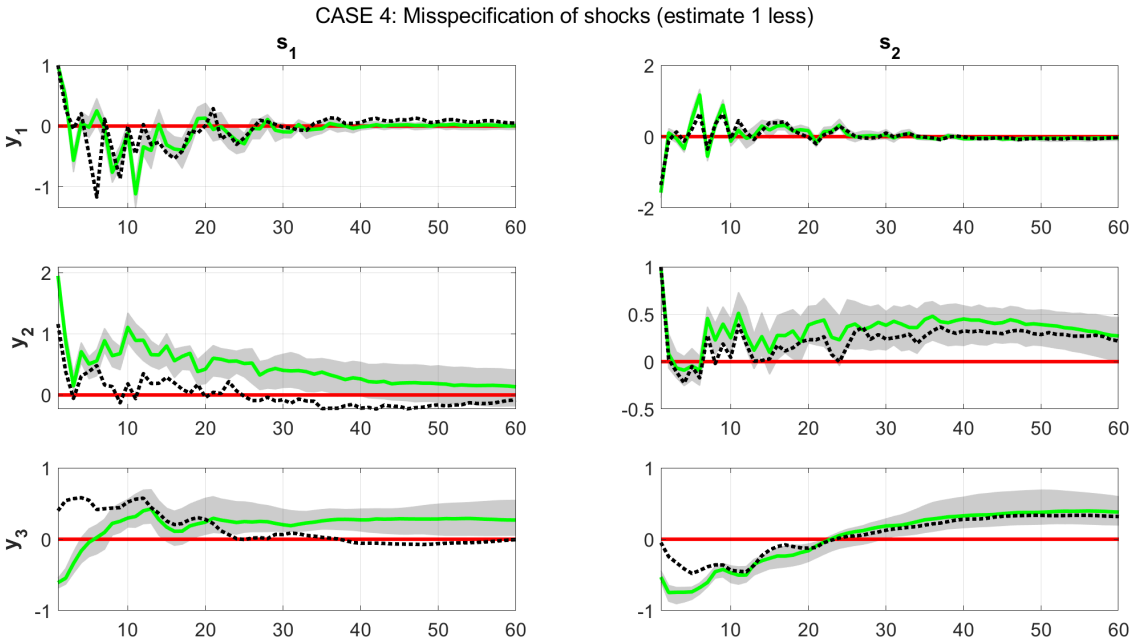


Figure C3: *Impulse response functions of the first three artificially generated variables (denoted as y_1, y_2, y_3) in response to the three identified shocks (denoted as s_1, s_2, s_3) in model C4 (misspecified number of shocks – one less). The green solid lines show the posterior median IRFs over the 500 Monte Carlo iterations, and the gray shaded areas their associated 90% bands. The true IRFs based on the DGP are shown using the black dashed lines.*

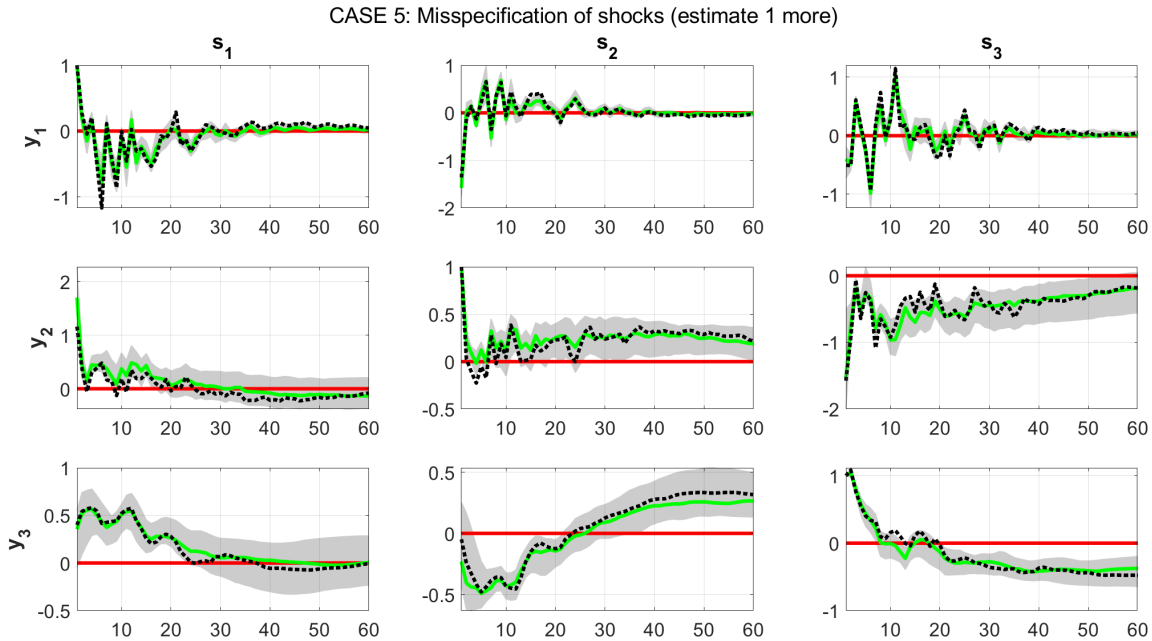
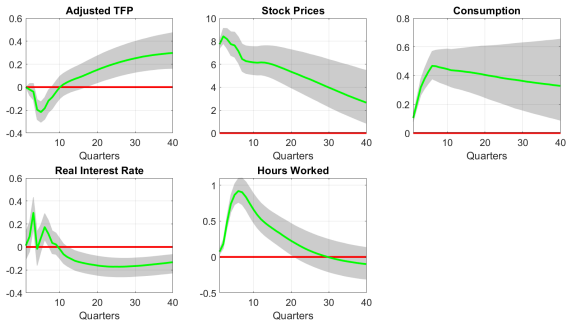


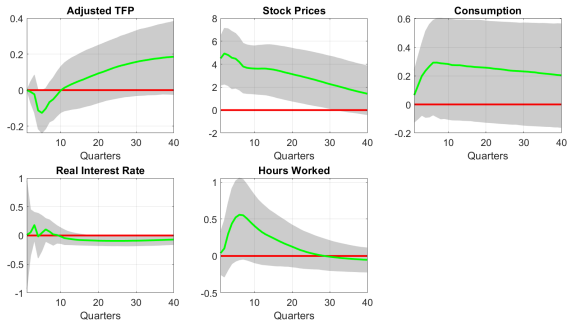
Figure C4: *Impulse response functions of the first three artificially generated variables (denoted as y_1, y_2, y_3) in response to the three identified shocks (denoted as s_1, s_2, s_3) in model C5 (misspecified number of shocks – one more). The green solid lines show the posterior median IRFs over the 500 Monte Carlo iterations, and the gray shaded areas their associated 90% bands. The true IRFs based on the DGP are shown using the black dashed lines.*

C.2 Additional empirical exercise: Measuring optimism shocks

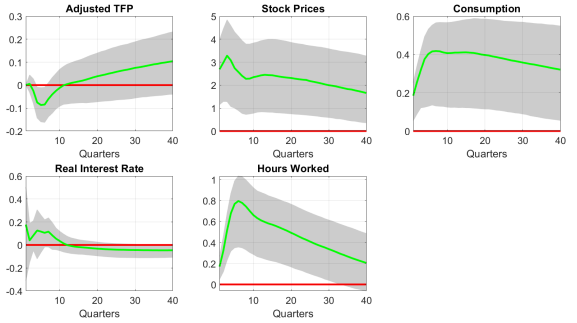
Here I undertake an additional empirical exercise that will help shed more light on the performance of the new algorithm in real data. This empirical exercise is based on Section 6 of Arias et al. (2018). These authors use the example in Beaudry et al. (2011) in order to compare their novel importance sampling algorithm to the penalty function approach (PFA) of Mountford and Uhlig (2009). For the sake of comparability, we maintain their empirical setting, and for that reason we estimate VAR(4) models using the following five dependent variables: adjusted TFP, stock prices, consumption, the real interest rate, and hours worked. We identify a single optimism shock by restricting contemporaneously TFP to have a zero response, and stock prices to react positively. The signs in the remaining three variables are not restricted. Panels (a) and (b) in Figure C5 show the estimated responses using the PFA and importance sampling algorithms, respectively. These two panels are identical to panels (a) and (b) in Figure 1 of Arias et al. (2018). The main point these authors make in their study is that the PFA algorithm ends up distorting the responses of stock prices, consumption and hours. Once their proposed importance sampling algorithm is considered, the significant responses found in Beaudry et al. (2011) disappear.



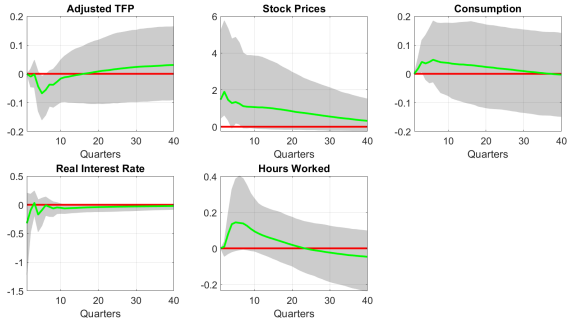
(a) Mountford and Uhlig (2009) algorithm



(b) Arias et al. (2018) algorithm



(c) Factor sign restrictions algorithm



(d) Factor sign restrictions algorithm, with additional zero restrictions

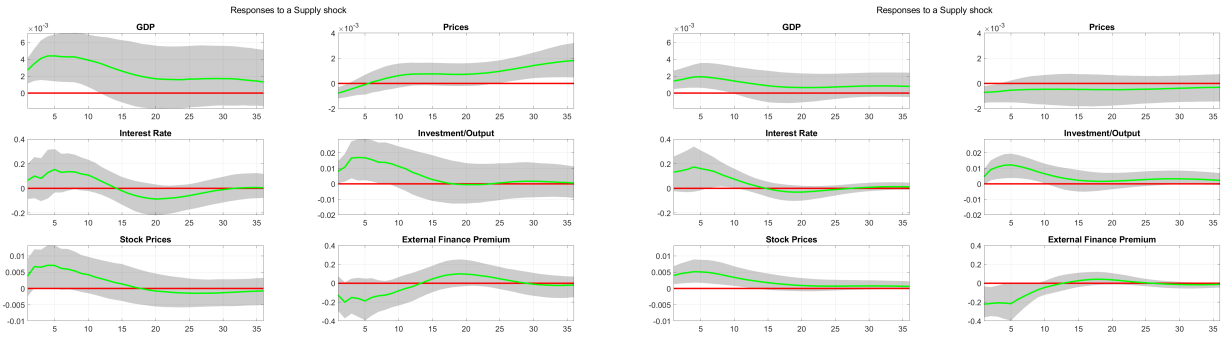
Figure C5: *This figure replicates the results in Beaudry et al. (2011), using a five variable VAR for assessing the effects of an optimism shock. The sign-restricted impulse responses of the five variables are estimated using (a) the PFA algorithm of Mountford and Uhlig (2009); (b) the importance sampling approach of Arias et al. (2018); (c) the Gibbs sampler algorithm proposed in this paper; and (d) the algorithm of this paper, with additional zero restrictions in consumption and hours.*

Panel (c) of Figure C5 shows the results from the factor sign restrictions algorithm using the same zero restriction on TFP and positive sign restriction on stock prices. The response of stock prices is not as pronounced as in Beaudry et al. (2011), and in general the responses for TFP, stock prices and real interest rate are equivalent to Arias et al. (2018). However, the responses of consumption and hours are still strongly different from zero, even though they have error bands and shapes that look much closer to those produced by the algorithm of Arias et al. (2018). Nevertheless, one aspect of the factor sign restrictions is that we can explicitly derive the implied fit to the VAR of imposing various restrictions. Therefore, we can explicitly test the premise of Arias et al. (2018) that consumption and hours are not affected by optimism shocks. Panel (d) in Figure C5 repeats estimation of the VAR using factor sign restrictions algorithm with additional zero restrictions in consumption and hours. The IRFs now look quantitatively and qualitatively closer to those in panel (b). Most importantly, we

are able to test whether the model in panel (c) or (d) is supported by the data, that is, test whether the zero restrictions in consumption and hours. The DIC for the model without these restrictions is -13109.49 while the DIC for the model with the two zero restrictions is -15267.64. Thus, data evidence (which is conditional, of course, on the specific parametric likelihood specification and prior) suggests that the premise of Arias et al. (2018) – that optimism shocks do not affect consumption and hours – is correct.

C.3 Additional results for the baseline VAR of Furlanetto et al. (2019)

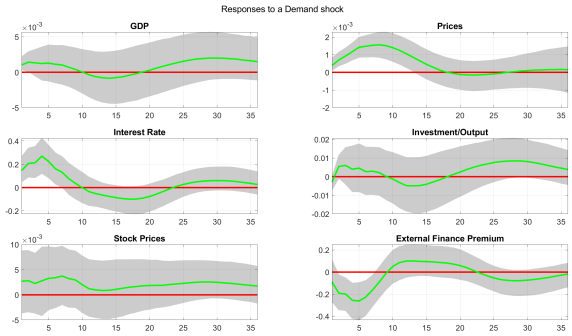
The main paper presents only the impulse responses to a financial shock using the baseline VAR specification of Furlanetto et al. (2019). Figures C6 - C9 illustrate that when looking at the four other macro shocks, results of the new algorithm and the Rubio-Ramírez et al. (2010) algorithm are qualitatively identical and quantitatively very similar.



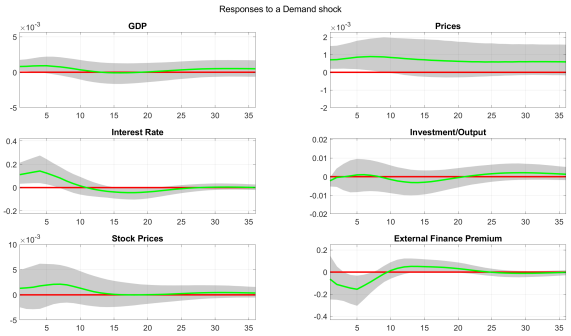
(a) Furlanetto et al. (2019) algorithm

(b) Factor sign restrictions algorithm

Figure C6: *This figure replicates the impulse response functions to an aggregate supply shock using the baseline specification of Furlanetto et al. (2019). Panel (a) shows results based on the exact configuration of Furlanetto et al. (2019, see Figure 1), using the algorithm of Rubio-Ramírez et al. (2010). Panel (b) replicates the same financial shock using the new sign restrictions algorithm.*

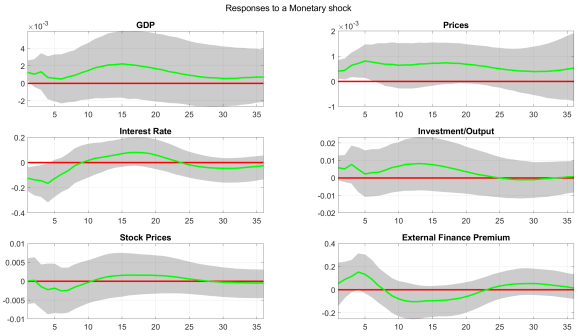


(a) Furlanetto et al. (2019) algorithm

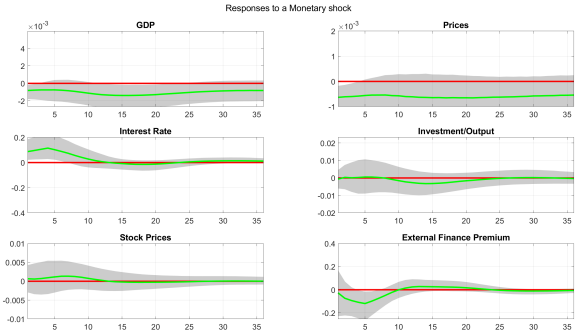


(b) Factor sign restrictions algorithm

Figure C7: *This figure replicates the impulse response functions to an aggregate demand shock using the baseline specification of Furlanetto et al. (2019). Panel (a) shows results based on the exact configuration of Furlanetto et al. (2019, see Figure 1), using the algorithm of Rubio-Ramírez et al. (2010). Panel (b) replicates the same financial shock using the new sign restrictions algorithm.*

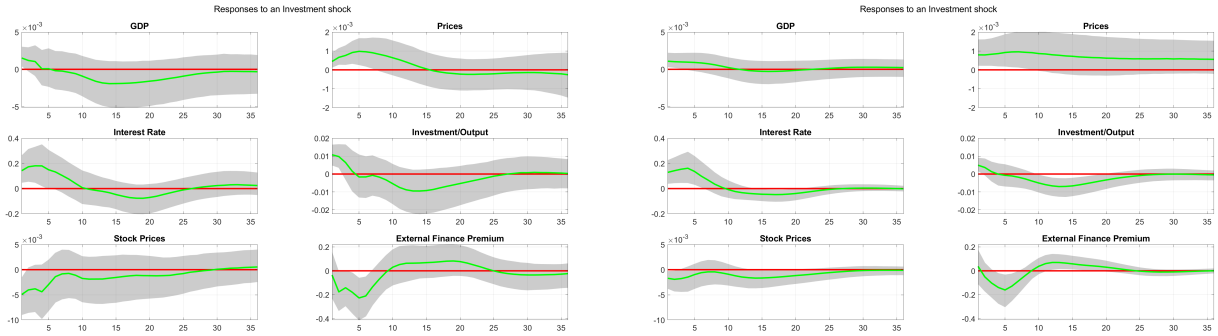


(a) Furlanetto et al. (2019) algorithm



(b) Factor sign restrictions algorithm

Figure C8: *This figure replicates the impulse response functions to a monetary policy shock using the baseline specification of Furlanetto et al. (2019). Panel (a) shows results based on the exact configuration of Furlanetto et al. (2019, see Figure 1), using the algorithm of Rubio-Ramírez et al. (2010). Panel (b) replicates the same financial shock using the new sign restrictions algorithm.*

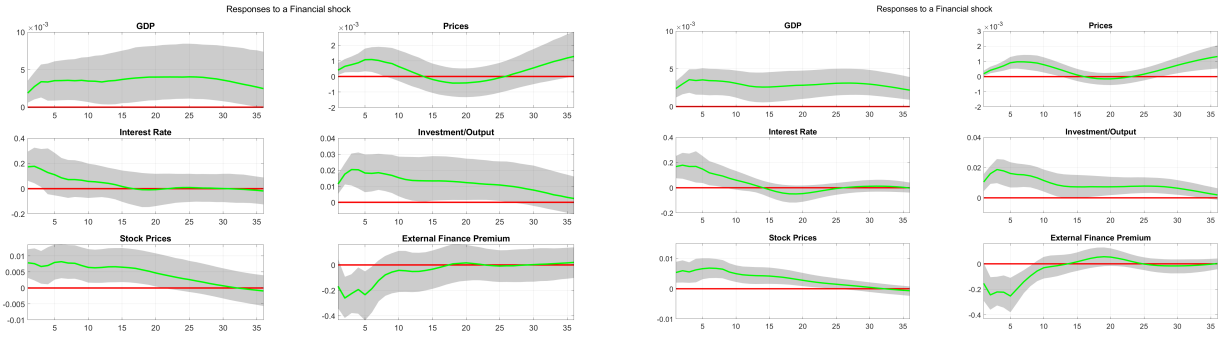


(a) Furlanetto et al. (2019) algorithm (b) Factor sign restrictions algorithm

Figure C9: *This figure replicates the impulse response functions to an investment shock using the baseline specification of Furlanetto et al. (2019). Panel (a) shows results based on the exact configuration of Furlanetto et al. (2019, see Figure 1), using the algorithm of Rubio-Ramírez et al. (2010). Panel (b) replicates the same financial shock using the new sign restrictions algorithm.*

C.3.1 Financial shock using a noninformative prior

Just for this specific exercisxe, it would be interesting to exchange the informative (shrinkage) horseshoe prior for a noninformative prior in order to bring the new methodology closer to the settings adopted by Furlanetto et al. (2019). These authors estimate the baseline VAR using diffuse prior distributions. Figure C10 plots these results and compared to Figure 2, the shapes of the IRFs are now identical. The IRFs in panel (a) still have wider bands, but this is to be expected since the covariance matrix in panel (b) is sampled from a parsimonious factor representation (and not from an inverse Wishart posterior).



(a) Furlanetto et al. (2019) algorithm (b) Factor sign restrictions algorithm

Figure C10: *This figure replicates the impulse response functions to an financial shock using the baseline specification of Furlanetto et al. (2019). This is identical to Figure 2*

C.4 Additional impulse responses from the large-scale, 15-variable VAR

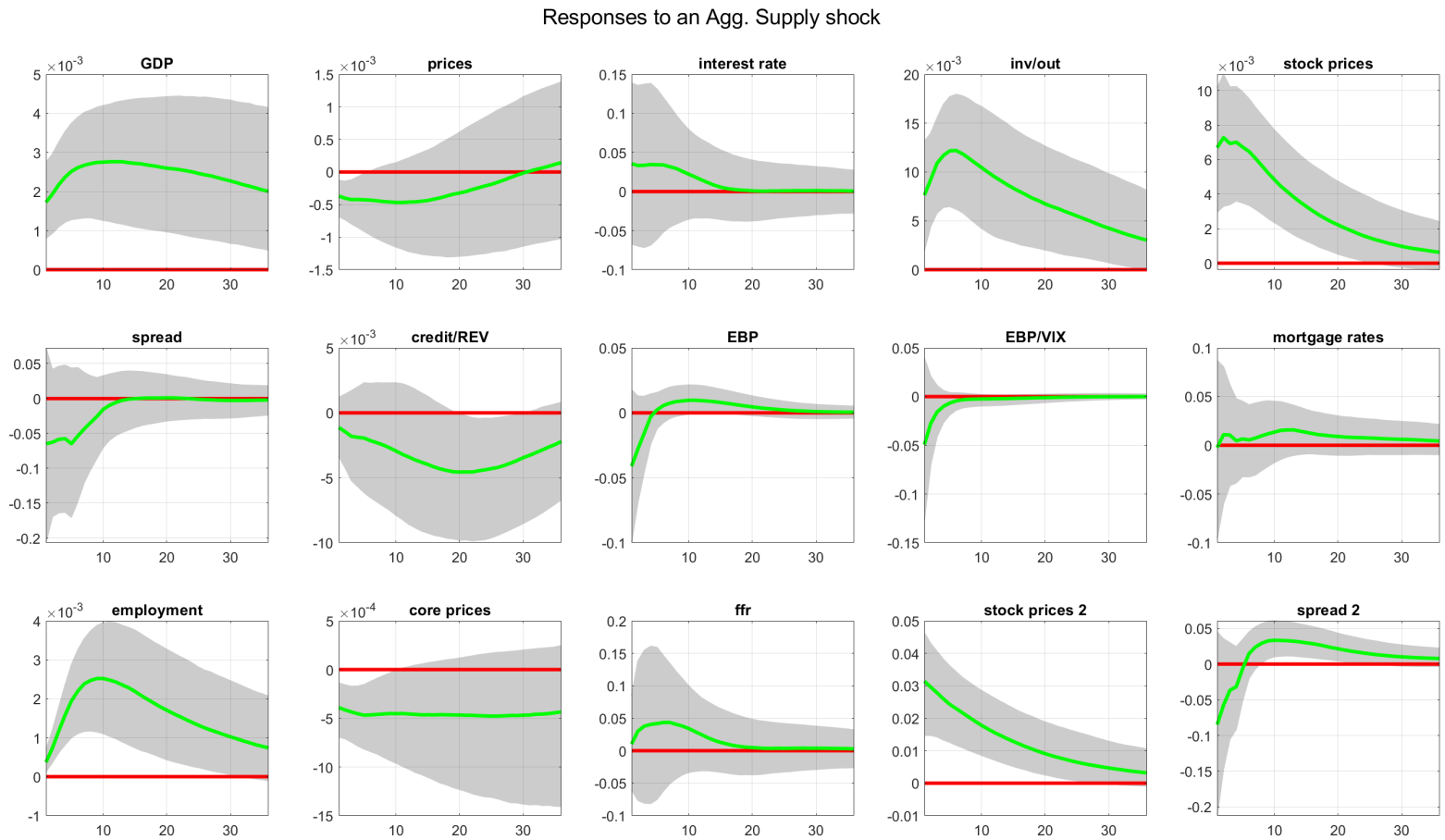


Figure C11: *Impulses response functions to an aggregate supply shock in the large, 15-variable VAR with seven shocks identified in total.*

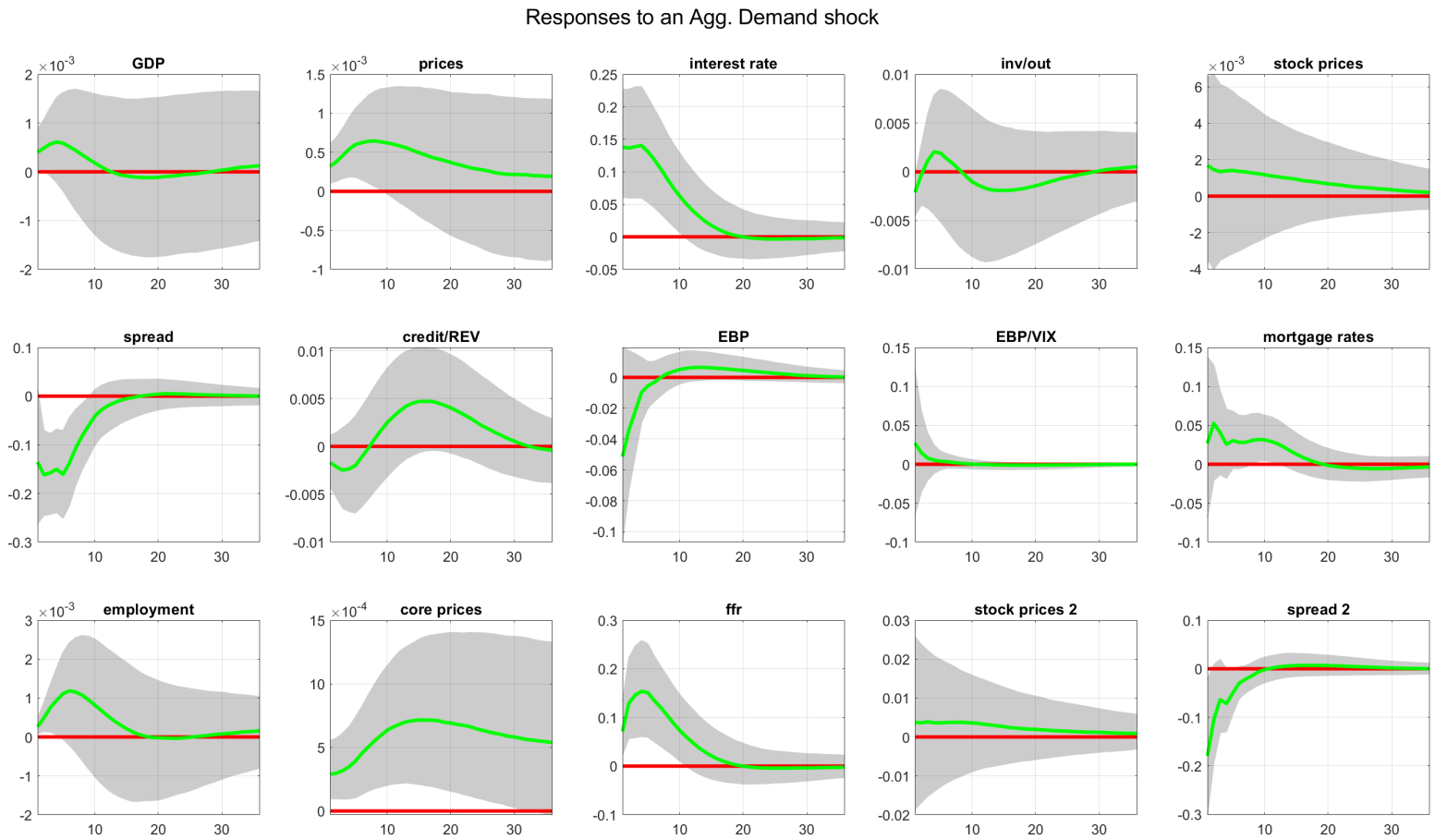


Figure C12: *Impulses response functions to an aggregate demand shock in the large, 15-variable VAR with seven shocks identified in total.*

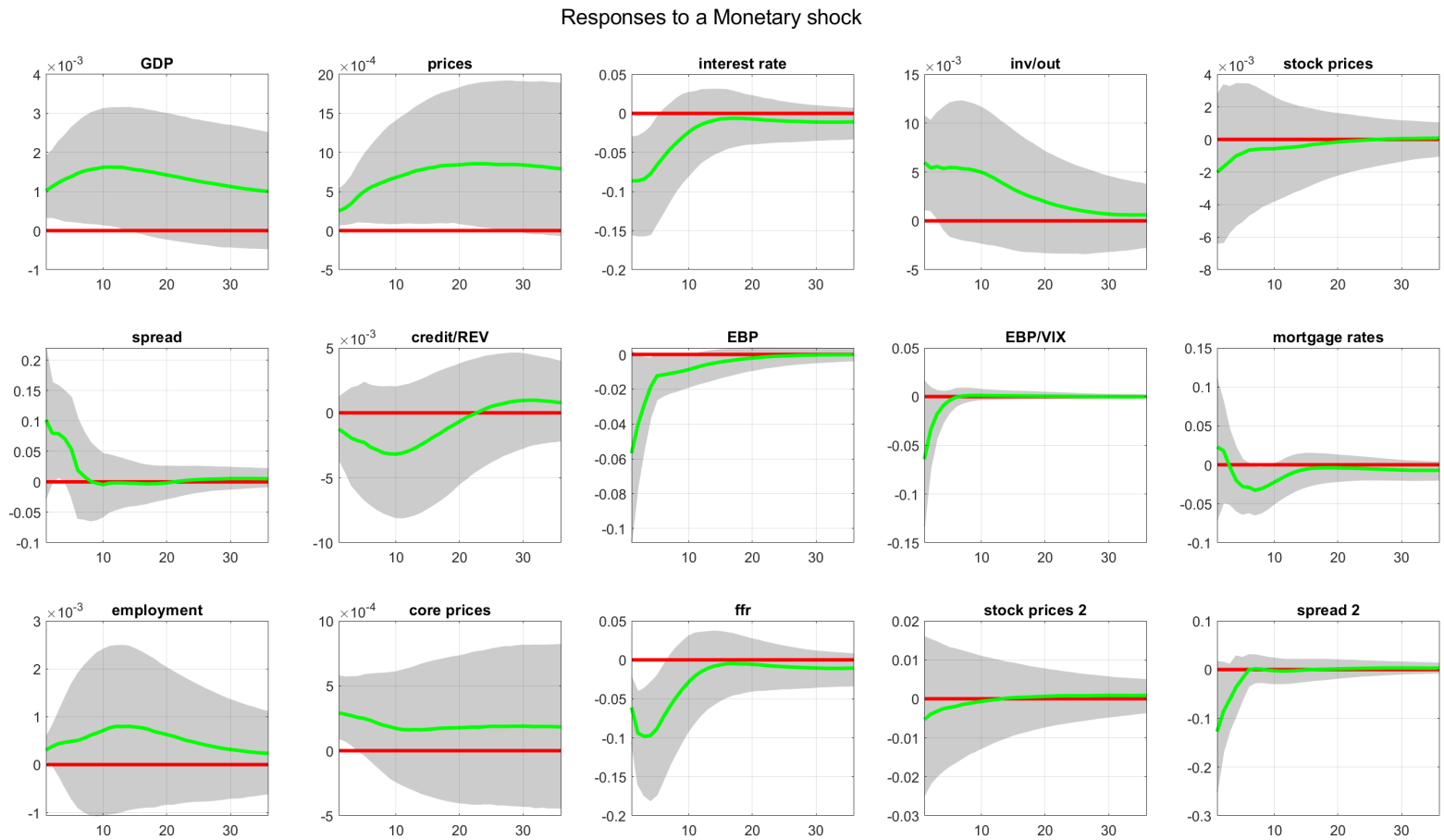


Figure C13: *Impulses response functions to a monetary policy shock in the large, 15-variable VAR with seven shocks identified in total.*

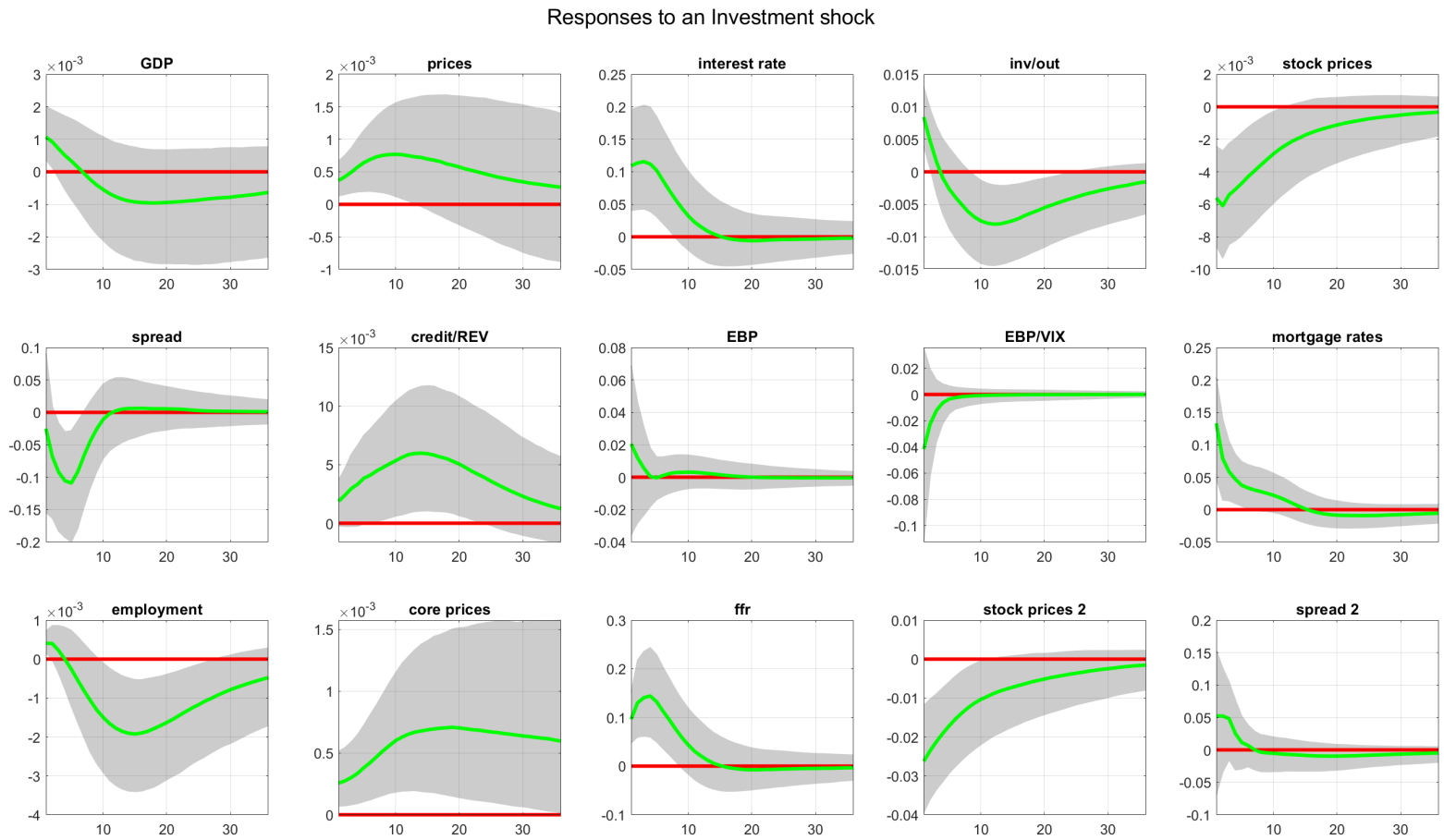


Figure C14: *Impulses response functions to an investment shock in the large, 15-variable VAR with seven shocks identified in total.*

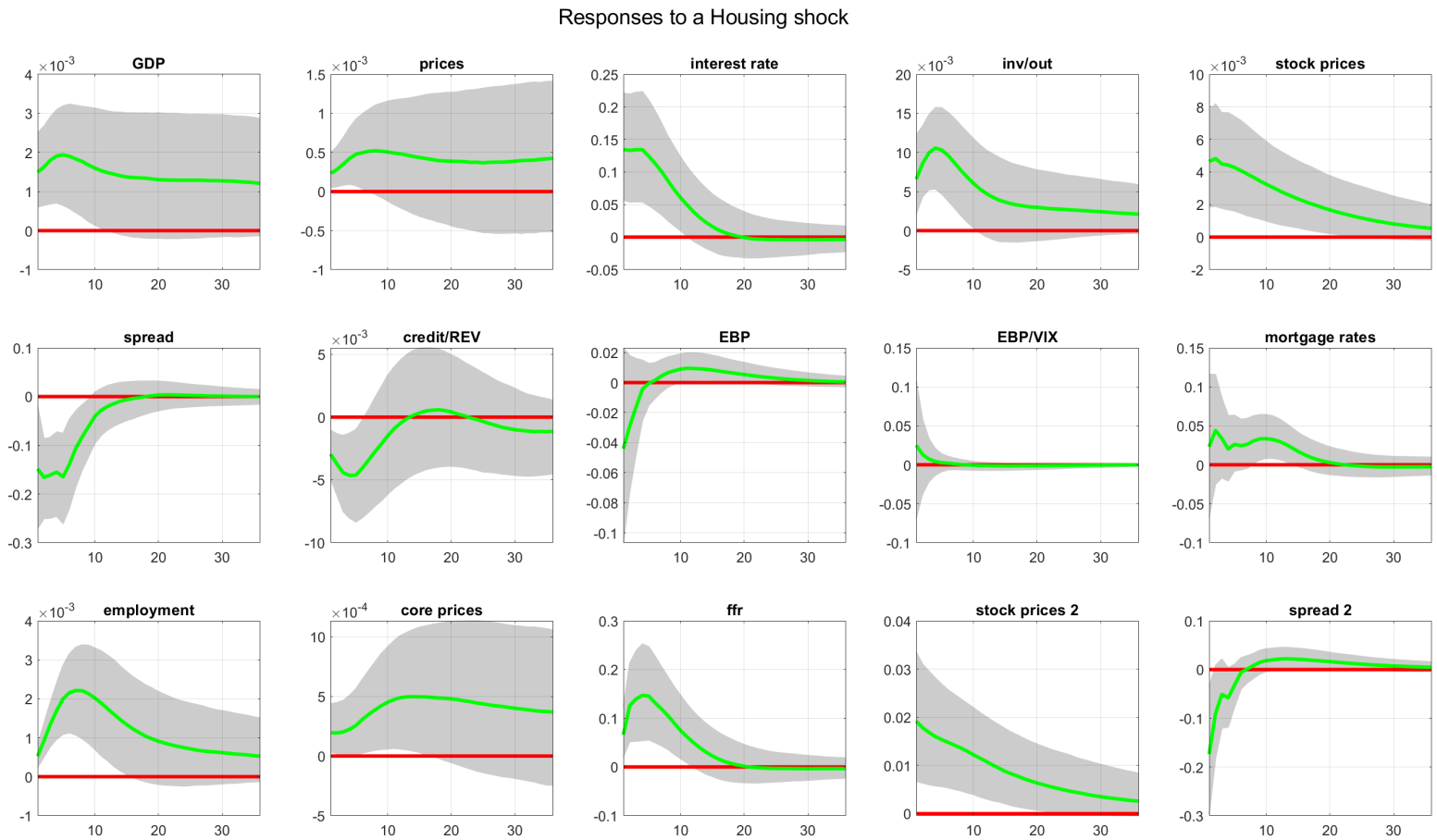


Figure C15: *Impulses response functions to a housing shock in the large, 15-variable VAR with seven shocks identified in total.*

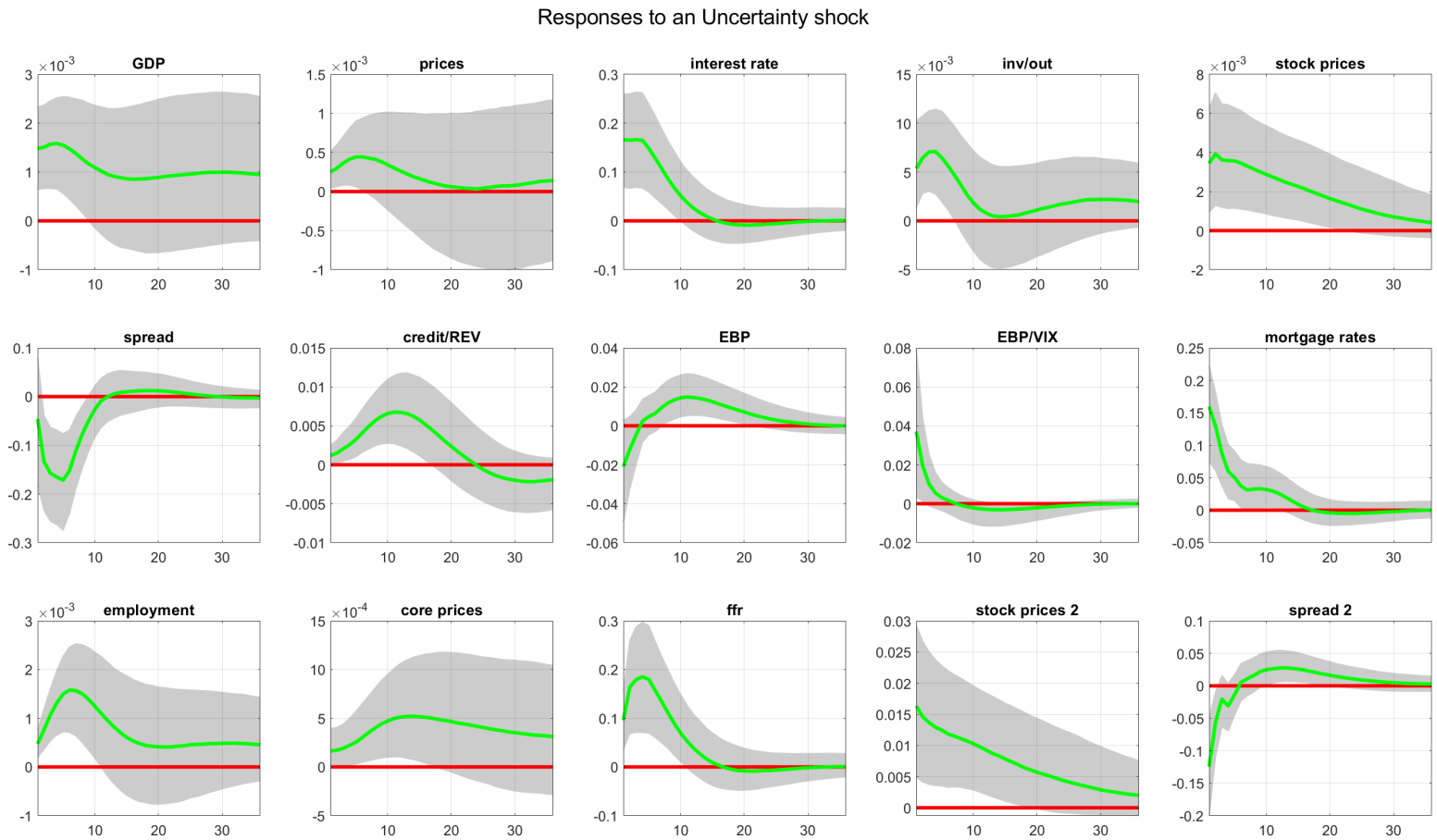


Figure C16: *Impulses response functions to an uncertainty shock in the large, 15-variable VAR with seven shocks identified in total.*